INTEGRATED PROJECT REPORT

Healthcare Data Analysis: From Data Strategy to Decision Support

A capstone that unites data sourcing and quality, statistical and predictive analysis, visualization and reporting, with a reflective self-assessment and career pathway

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Code, data and portfolio index: https://github.com/ipshitav71-wq/healthcare-data-analysis-capstone

> Important note: the case study in this report is hypothetical. All patient-level data were simulated by the author to resemble routine hospital discharge records. No real patients, hospitals or organisations are represented, and the numbers must not be used as real-world estimates. The value of the report lies in the integrated method, the reasoning that connects each step to a decision, and the honest assessment of what worked and what did not.

Executive Summary

Purpose. This final report brings together four earlier pieces of work into one project: a data collection and quality strategy, a data visualization plan, a statistical analysis and predictive modeling framework, and an analytical report. Instead of presenting them as separate documents, it follows one case from start to finish: a four-hospital network that wants to understand and reduce 30-day unplanned readmissions.

What the integrated project found. Across the network the readmission rate was steady (14.5% in the first 18 months and 14.5% in the last six), but one hospital (Hospital D) rose from 15.0% to 18.0%. In the same hospital, completion of follow-up within 7 days fell from 55.9% to 30.4%. Patients without early follow-up were readmitted at 20.0% compared with 10.0% for those with it (adjusted OR 0.45). A risk model with modest accuracy (AUROC 0.644) helped prioritise follow-up calls, capturing 33.6% of readmissions in the highest-risk 20% of discharges, while a fairness audit showed the model under-predicting risk for rural patients (observed/expected ratio 1.24).

Why integration mattered. Each finding depended on more than one component. The Hospital D problem was visible only because quality checks produced a trustworthy dataset, the analysis was done at hospital level, and a control chart made the signal easy to see. A balanced-class-weight model looked fine on ranking but produced misleading risk numbers, which only a calibration check exposed. Without a plan for communication, none of this would have reached the people who can act.

Summary of unified recommendations (Section 9)

Area	Recommendation	Evidence
Operations	Restore 7-day follow-up at Hospital D to about 56% and make medication reconciliation universal	Sections 5.2, 5.3
Analytics	Monitor readmissions by hospital and ward with control charts; pilot the risk model in silent mode, then use it only to prioritise calls	Sections 5.2, 5.4
Equity	Close the rural and low-income follow-up gap; report equity measures every quarter; recalibrate the model for under-predicted groups	Sections 5.4, 5.5

Area	Recommendation	Evidence
Data	Adopt data-quality gates and a documented source log before any analysis; reduce missing income, comorbidity and medication fields	Section 4
Communication	Deliver an accessible dashboard and one-page briefing with text descriptions of every chart	Section 6
Governance	Obtain ethics approval, keep humans in control of decisions, and review model drift and fairness monthly	Section 7

Learning and next steps. Appendix A is a reflective self-assessment. Key lessons: design from the decision, not from the tool; describe every visual in words as well as showing it; give concrete, worked examples; test the common pitfalls (such as class imbalance and leakage) rather than only naming them. The main gaps are hands-on experience with real, messy data and with building dashboards in Power BI and SQL, which the career pathway in Appendix A.6 addresses.

1. Introduction and Project Overview

1.1 Purpose and Scope

The purpose of this report is to show an integrated understanding of healthcare data analysis. Real projects rarely fail because a single technique was applied badly; they fail at the joins between stages: data that was never checked, a model that was never explained, a chart that nobody could read, a recommendation not connected to evidence. The report therefore treats the whole workflow as one system and shows, with a single running case, how each stage depends on the others.

1.2 The Five Components

Component	Earlier deliverable	Role in this integrated project
1. Data strategy	Data collection and quality strategy: public sources, selection criteria, acquisition plan	Decides what data to use, how to obtain it, and under what conditions (Section 4.1)
2. Data quality	Quality framework, validation and cleaning plan, risk assessment	Turns raw records into a trustworthy cohort and records every exclusion (Sections 4.2, 4.3)
3. Analysis and modelling	Statistical analysis and predictive modeling framework	Finds trends and drivers and builds a validated, fair risk model (Section 5)
4. Visualization	Data visualization plan with dashboard mock-ups	Turns results into accessible visuals for each audience (Section 6)
5. Reporting	Analytical report with recommendations	Links evidence to decisions and states limitations (Sections 5 to 9)

1.3 The Unified Case

A regional network of four hospitals asked three questions: Is the 30-day readmission rate changing? Where are the problems and which patients are at risk? What should the network do about it? The case uses 24,268 adult discharges (16,968 patients) from September 2024 to August 2026, generated by simulation so that every step could be carried out and checked without real patient data. Appendix C lists the repositories holding the code and data.

1.4 How This Report Meets the Task Requirements

Requirement	Where addressed
Review of previous outputs and key insights	Section 2 (insight table K1 to K8)
Unified structure that logically brings elements together	Section 3 (framework) and the section order of this report
Self-assessment of approach, methods and growth	Appendix A (A.1 to A.7)
Where strategies interlink and integrated efforts improve outcomes	Section 8
Future recommendations, pitfalls and lessons learned	Section 10
Best practices summary	Section 10.1 and 10.3
Critical assessment and career development pathway	Appendix A.3 and A.6

2. Review of Previous Work: Key Insights Carried Forward

The earlier deliverables were reviewed to decide what to carry into the integrated project. The table lists the main insights (K1 to K8) and how each is used here.

ID	Source work	Key insight	How it is used in this report
K1	Data strategy	Choose datasets by relevance, provenance, documentation, currency, licence and reproducible access; log the source and version	Source scoring and data-source log (Section 4.1)
K2	Data quality	Check completeness, correctness, concordance, plausibility and currency (Weiskopf & Weng, 2013)	Quality checks with actual results (Section 4.2)
K3	Data quality	Anticipate risks: missing data, duplicates, coding inconsistency, bias, drift, re-identification	Case risk register (Section 4.3)
K4	Visualization	Design from decisions and personas; use a chart guide; make visuals accessible	Insight-to-visual map and dashboard (Section 6)
K5	Visualization	Power BI for dashboards, Python for statistical charts, PostgreSQL for modelling	Tool roles and model sketch (Section 6.4)
K6	Statistics and modelling	Split data by patient and date; prevent leakage; use nested validation, calibration and fairness audits	Analysis design (Section 5)
K7	Reporting	Tie every recommendation to evidence; give worked examples; describe every visual in text	Matrix and recommendations (Sections 5.6, 9)
K8	All	Ethics and privacy are checks at every stage, not a final review	Lifecycle ethics table (Section 7)

Feedback on earlier submissions also shaped this report. Evaluators asked for concrete examples, for mock-ups to be described rather than only drawn, for practical handling of class imbalance, and for an end-to-end case study. These points are addressed throughout and are itemised in Appendix A.2.

3. Unified Framework and Project Structure

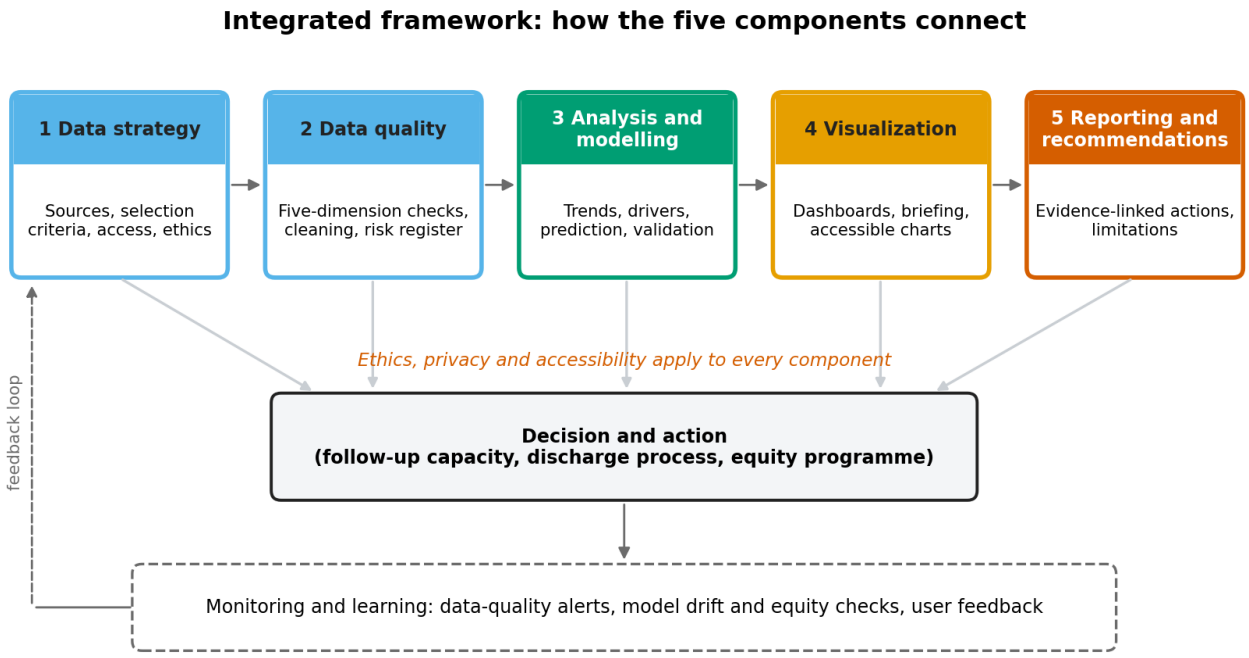


Figure 1. Integrated framework connecting the five components

Figure in words: Five boxes run left to right: data strategy, data quality, analysis and modelling, visualization, and reporting and recommendations. Arrows from every box lead to a central “Decision and action” box, which leads to a “Monitoring and learning” box; a dashed arrow returns from monitoring to the start. A note states that ethics, privacy and accessibility apply to all components. Take-away: the work is a loop, not a line, and every stage feeds the decision.

Stage	Key question	Main outputs	Quality gate before moving on
Data strategy	What data answers the question, and may we use it?	Source log, selection scores, access and licence record	Purpose, permissions and documentation confirmed
Data quality	Can the data be trusted?	Validation report, cleaning log, cohort flow, risk register	Critical checks pass or exceptions documented
Analysis and modelling	What changed, why, and who is at risk?	Trend, driver and model results with uncertainty; fairness audit	Test data untouched; calibration and subgroup checks done
Visualization	How will each audience understand and act?	Dashboard, briefing, text descriptions, accessibility check	Each visual linked to a decision; accessibility checklist passed
Reporting	What should we do, and how sure are we?	Evidence-linked recommendations, limitations	Every recommendation traced to evidence
Monitoring	Is it still working, and for everyone?	Drift, equity and quality dashboards	Review dates and owners agreed

4. Data Strategy and Quality in the Case

4.1 Choosing the Data (Component 1)

The data strategy lists the public sources considered for healthcare analytics: government portals (healthdata.gov, CDC Open Data, CMS Provider Data Catalog), international repositories (WHO Global Health Observatory), credentialed clinical databases (MIMIC-IV on PhysioNet), claims and utilisation data (HCUP), and curated learning datasets (Kaggle and UCI). Candidate datasets for the readmission question were scored against the selection criteria (1 = weak, 3 = strong) using the author's judgement.

Criterion	UCI Diabetes 130-US Hospitals	MIMIC-IV (ICU)	CMS hospital quality data	Simulated cohort (used)
Relevance to readmission question	3 (has readmission label)	2 (ICU stays, readmission derivable)	2 (hospital-level, not patient-level)	3 (designed for the question)
Provenance and documentation	3 (published paper and codebook)	3 (peer-reviewed descriptor)	3 (government source)	3 (code and dictionary provided)
Currency	1 (1999–2008)	2 (recent releases)	3 (regularly updated)	3 (to Aug 2026)
Access and licence	3 (open)	1 (training and agreement needed)	3 (open)	3 (own data)
Privacy risk	2 (de-identified)	2 (de-identified, restricted use)	3 (aggregate)	3 (no real patients)
Allows the full workflow (patient-level, dates, follow-up)	2 (no dates, no follow-up variable)	2	1	3
Total (out of 18)	14	12	15	18

Table 1. Source scoring. The simulated cohort scores highest because it was built for the question; for a real replication, the UCI dataset is the best open practice option and MIMIC-IV the most realistic once credentialed.

Decision: a simulated cohort was used because the project is hypothetical and because it allowed every stage, including follow-up completion and dates, to be demonstrated without any privacy risk. A source log (name, version, date, licence, access method, checksum) records each file.

4.2 Checking the Data (Component 2)

The five-dimension quality framework was applied as a set of concrete checks. Results are from running the checks on the case data.

Dimension	Check applied	Result in the case data
Completeness	Percentage missing per field after cleaning	Income group 6.8%, comorbidity count 2.5%, medication reconciliation 4.1%; length of stay missing in 199 records, which were excluded
Uniqueness	Duplicate encounter identifiers	153 duplicate rows found in the raw extract of 25,583 and removed; 0 remain
Correctness	Range and category checks	Age 18–96 years, length of stay 1–38 days, comorbidities 0–7, prior admissions 0–7; 0 values outside limits; all ward and discharge categories valid; all yes/no fields valid

Dimension	Check applied	Result in the case data
Concordance	Consistency of linked fields	Age, sex and residence agree across the 5,894 patients with several admissions (0 conflicts); excluded categories (deaths, hospice, transfers) leave 0 records
Plausibility	Review of distributions and subgroup patterns	No impossible combinations found; missing income was more common among rural patients (11% vs 5.1%), so it is treated as informative, not random
Currency	Coverage window and outcome follow-up	Discharges to 31 Aug 2026 with 30 days of follow-up complete before the 24 Oct 2026 extract

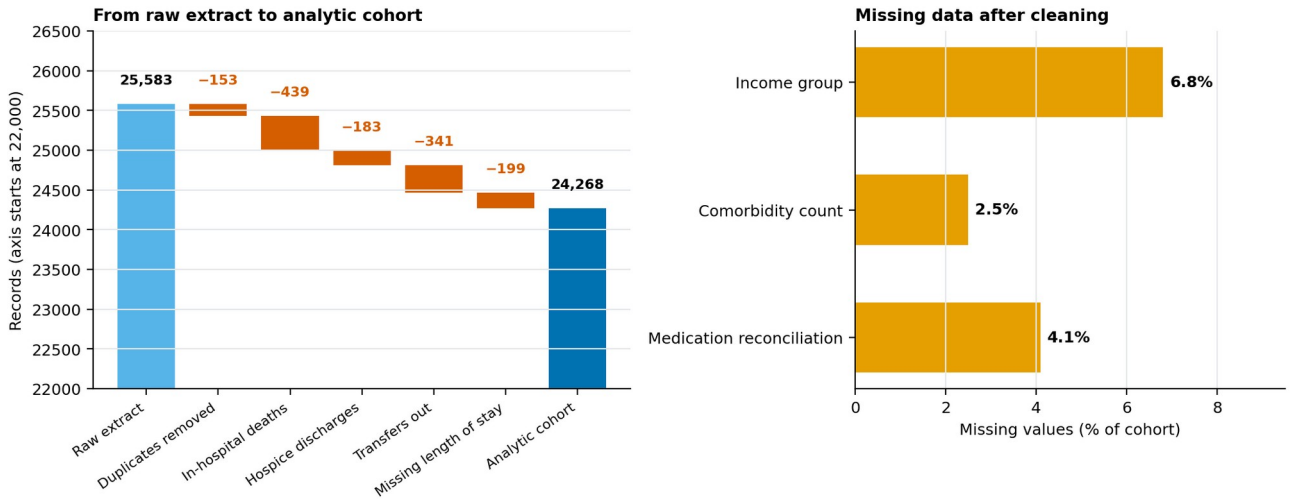


Figure 2. Data quality at a glance: cohort flow (left) and remaining missing data (right)

Figure in words: Left: the raw extract holds 25,583 records. Removing 153 duplicates, 439 in-hospital deaths, 183 hospice discharges, 341 transfers and 199 records without length of stay leaves 24,268 discharges. Right: after cleaning, 6.8% of income values, 2.5% of comorbidity values and 4.1% of medication reconciliation values are missing. Take-away: the cohort lost about 5% of records and income is the field needing most attention.

4.3 Risk Register for the Case

Risk	Where it appeared in the case	Mitigation applied	Residual risk
Duplicate records inflate counts	153 duplicate encounters in the raw extract	De-duplicate on encounter identifier; log the count	Low
Informative missing data	Income missing more often for rural patients	Keep “unknown” category; missing indicator for comorbidity; sensitivity check	Medium
Data leakage	Follow-up completion is recorded after discharge	Excluded from the prediction model; used only for explanation	Low
Class imbalance	Readmission is the minority outcome (14.5%)	Metrics beyond accuracy; calibration checked; re-weighting rejected after testing	Low
Hidden local problems	Network average stable while Hospital D rose	Hospital-level control charts and drill-down	Medium
Model drift	Hospital D observed/expected ratio above 1	Monthly calibration and subgroup monitoring; retraining triggers	Medium

Risk	Where it appeared in the case	Mitigation applied	Residual risk
Incomplete outcome capture	Only readmissions within the network are observed	Stated as a limitation; link to regional records in future	Medium
Re-identification and consent	Not applicable to simulated data; essential for real data	De-identification, access control, ethics approval, small-number suppression	Low if controls kept

5. Analysis and Modelling in the Case (Component 3)

5.1 Analytical Design

The eleven-step modeling workflow from the earlier framework was followed: frame the decision, define the cohort and outcome, split data by patient and date, pre-process, select variables, fit and tune models, evaluate, validate and audit for bias, document, deploy with oversight, and monitor. The outcome was unplanned 30-day readmission. The prediction model used only information known at discharge, trained on September 2024 to February 2026, tuned on March to May 2026, and tested once on June to August 2026.

Question	Method	Why
Is the rate changing?	p-chart with baseline limits and run rule; chi-square test of recent vs earlier periods	Separates real change from normal variation
What drives readmission?	Logistic regression with errors clustered by patient	Adjusts for patient mix; accounts for repeated admissions
Can we predict risk?	Penalised logistic regression vs gradient boosting; temporal split	Transparent baseline compared with a flexible model
Is the model fair?	Sensitivity and observed/expected ratio by subgroup	Detects groups the model serves poorly
What could be gained?	Scenario calculation from the fitted model	Order-of-magnitude benefit with stated assumptions

5.2 Trend: A Stable Network Hides a Rising Hospital

The network rate stayed near the 14.4% baseline with no month outside control limits (overall trend OR per month 1.001, p = 0.687). Hospital D, however, rose from 15.0% to 18.0% (p = 0.012) while Hospitals A to C moved from 14.4% to 13.6%. In August 2026 Hospital D's rate of 23.5% exceeded its upper control limit of 21.3%.

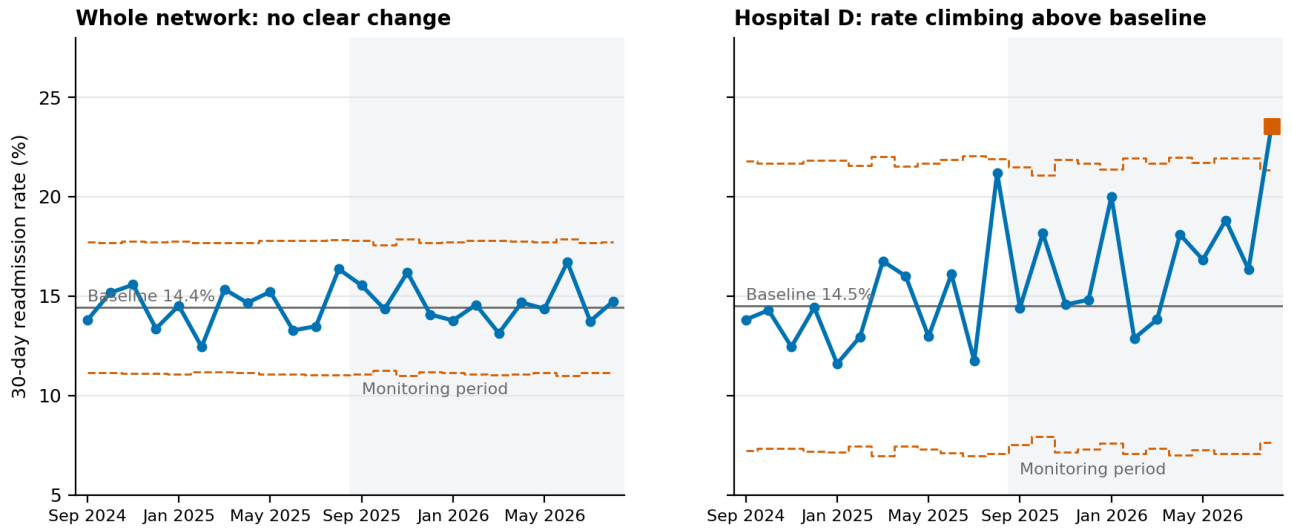


Figure 3. Control charts: whole network (left) and Hospital D (right)

Figure in words: Left: network monthly rates range from 12.4% to 16.7% inside limits of about 11.1% to 17.7% around 14.4%. Right: Hospital D swings more because monthly numbers are small; 5 of the last 6 months are above its 14.5% baseline and August 2026 (23.5%) is above the upper limit. Take-away: look at the level where action is possible.

5.3 Drivers: Follow-Up Is the Strongest Controllable Factor

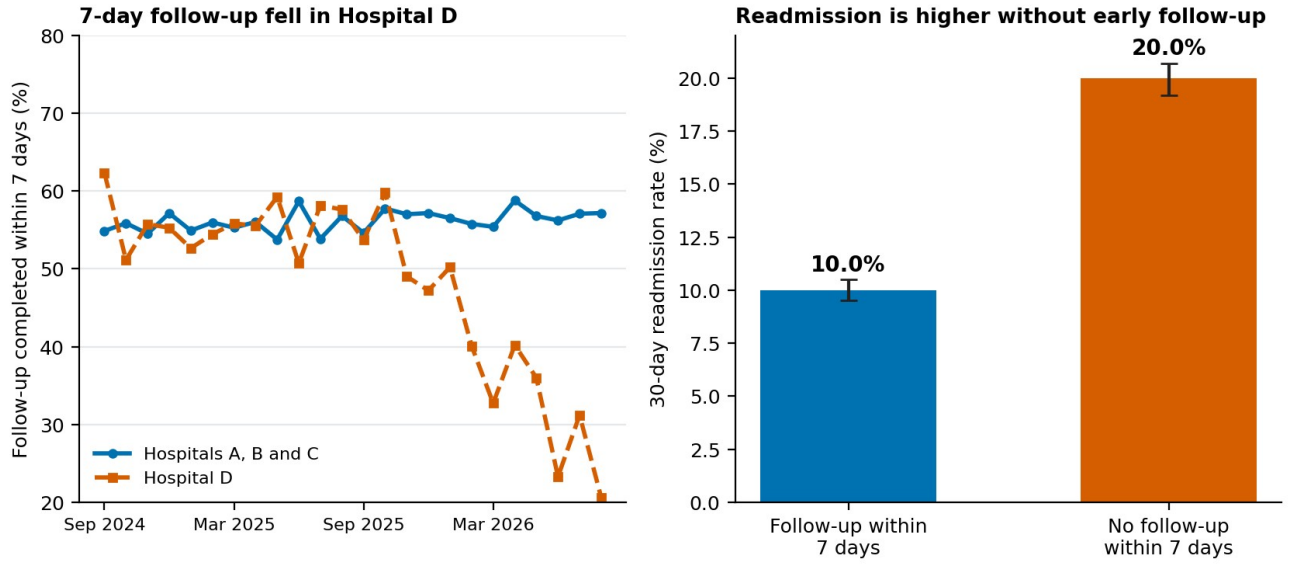


Figure 4. Follow-up within 7 days by month (left) and readmission by follow-up status (right)

Figure in words: Left: follow-up stays near 56.9% in Hospitals A to C but in Hospital D falls from 55.9% to 30.4%. Right: readmission is 10.0% with follow-up and 20.0% without. Take-away: the rise in Hospital D coincides with missed follow-up.

Factor	Adjusted OR	95% CI
Follow-up within 7 days	0.45	0.42 – 0.48
Medication reconciliation done (vs not)	0.75	0.68 – 0.81
Length of stay over 7 days	1.47	1.31 – 1.65

Factor	Adjusted OR	95% CI
Rural residence	1.33	1.23 – 1.44
Weekend discharge	1.25	1.15 – 1.36
Prior admissions (per additional)	1.23	1.18 – 1.28
Comorbidities (per additional)	1.21	1.17 – 1.25
Hospital D vs A	1.03	0.93 – 1.15

Table 2. Selected adjusted odds ratios for 30-day readmission (full table in the companion analytical report).

Adding follow-up to a model of the time trend in Hospital D removed the trend (shrinking the coefficient by roughly 100%), and Hospital D showed no excess risk after adjustment (OR 1.03). The interpretation is that a process change, not patient mix, explains the rise. Because the data are observational, this is a strong association that justifies action, not proof of cause.

5.4 Prediction: Useful for Prioritising, Not for Deciding

Measure (test period)	Result
AUROC (logistic regression)	0.644 (95% CI 0.619–0.674); gradient boosting 0.646, no real gain
Calibration slope / intercept	1.05 / 0.04
Top 20% flagged	Sensitivity 33.6%, positive predictive value 26.3%, 1.76 times better than random
Class re-weighting test	Same AUROC (0.644) but average predicted risk 47.8% against 15.0% observed; rejected

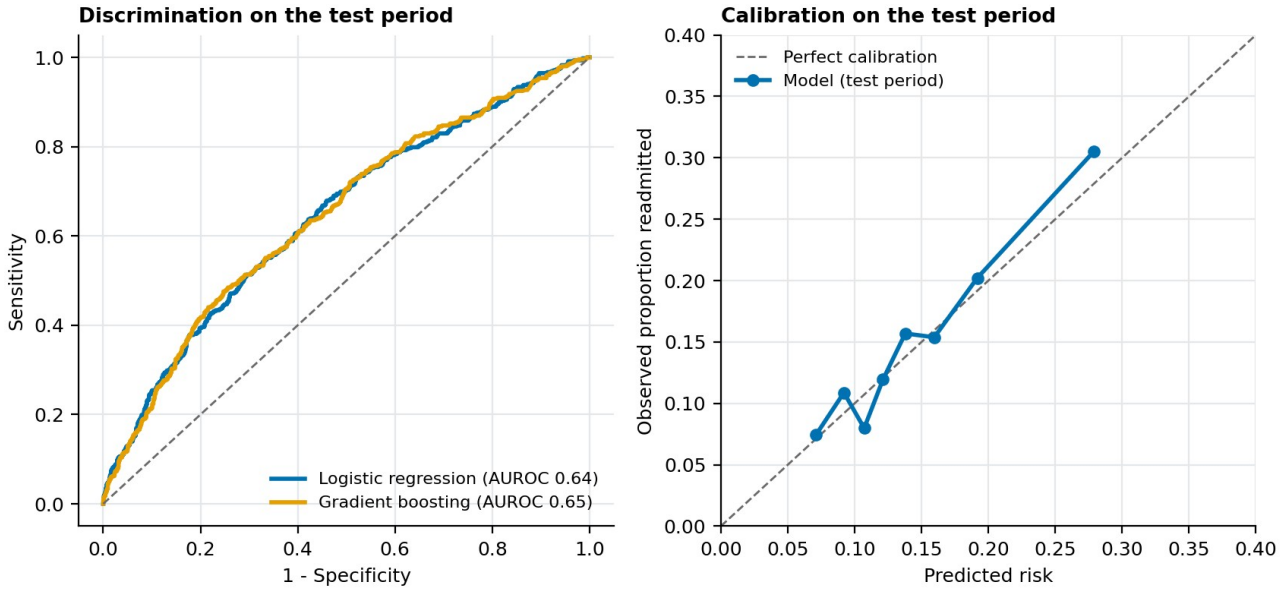


Figure 5. Discrimination (left) and calibration (right) on the test period

Figure in words: Left: both models rank patients modestly better than chance (AUROC 0.644 and 0.646). Right: predicted and observed risks agree closely. Take-away: the risk numbers are trustworthy overall, but ranking power is modest.

5.5 Fairness Audit

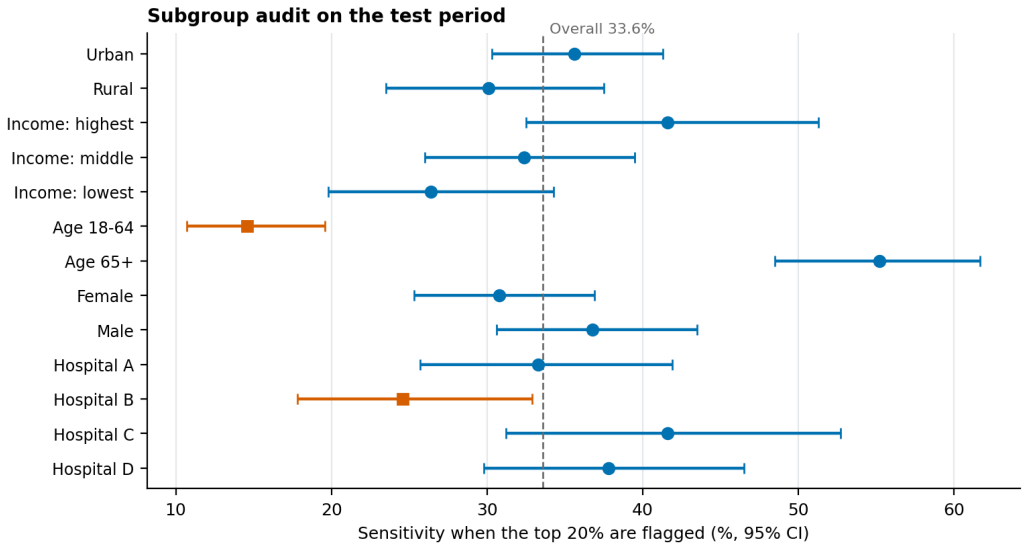


Figure 6. Sensitivity of the top-20% flag by subgroup

Figure in words: Sensitivity was 35.6% for urban and 30.1% for rural patients, 41.6% for the highest and 26.4% for the lowest income group, against 33.6% overall. The model under-predicted rural risk (observed/expected 1.24) and Hospital D risk (1.28). Most intervals overlap, so these are signals to monitor.

5.6 Evidence-to-Recommendation Matrix

Insight	Evidence	Recommendation
Network stable, Hospital D rising	Figure 3; Section 5.2	Escalate Hospital D; monitor by hospital and ward
Follow-up halves readmission and has fallen in Hospital D	Figure 4; Table 2	Restore follow-up capacity; target rural and low-income patients
Medication reconciliation protective and lower in Hospital D	Table 2; 66.1% vs 76.9%	Make reconciliation standard work
Model modest but useful for ranking	Section 5.4; Figure 5	Pilot as a prioritisation aid with human oversight
Rural and Hospital D under-prediction	Figure 6	Monitor equity; recalibrate

Scenario estimate: restoring follow-up at Hospital D to its earlier level would avert roughly 37 to 65 readmissions a year (about 325 bed-days at an assumed 5 days each), under the assumption that the follow-up effect is partly or fully causal.

6. Visualization and Communication (Component 4)

6.1 From Insight to Visual to Audience

Insight	Best visual (chart guide)	Primary audience	Decision supported
Hospital D is drifting above normal	Control chart with baseline and limits	Executives, hospital directors	Escalate and resource Hospital D

Insight	Best visual (chart guide)	Primary audience	Decision supported
Follow-up fell only in Hospital D	Two-line chart with target line	Hospital directors	Staffing for follow-up team
Follow-up halves readmission	Two-bar chart with intervals; “out of 100 patients” icon version for the public	All audiences	Justify investment
Who to call first	Ranked priority list with risk and status symbol	Follow-up team	Daily workload
Who is missed	Dot plot of sensitivity by group	Quality and equity leads	Equity programme

6.2 Integrated Dashboard Populated with Case Results

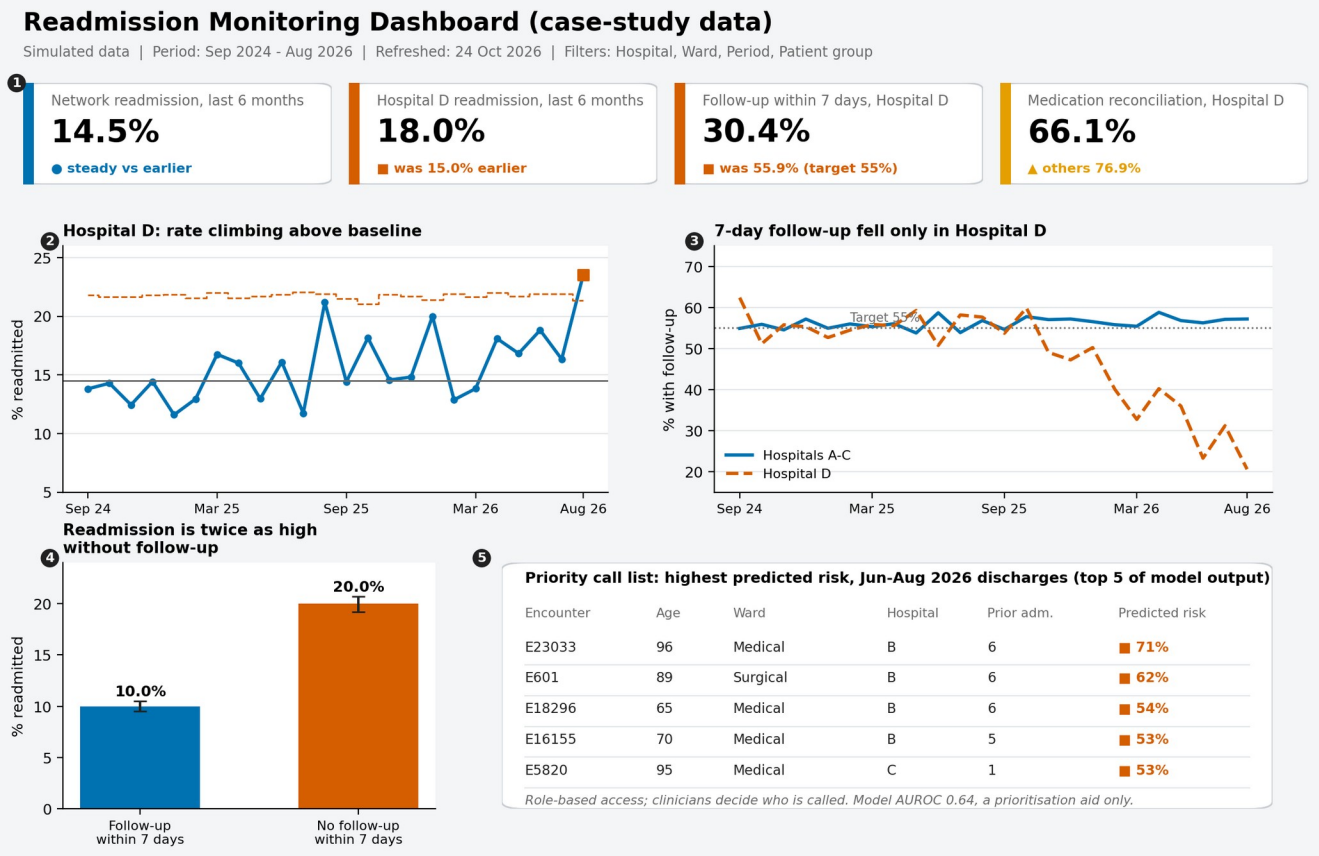


Figure 7. Integrated monitoring dashboard mock-up using the case-study results

Figure in words: Top row, four tiles with status symbols: network readmission 14.5% (steady), Hospital D readmission 18.0% (was 15.0%), Hospital D follow-up 30.4% (target 55%), medication reconciliation in Hospital D 66.1% (others 76.9%). Middle row: Hospital D control chart (left) and follow-up by month for Hospital D versus the other hospitals (right). Bottom row: readmission with versus without follow-up (left, 10.0% vs 20.0%) and a priority call list of the five highest predicted risks in June to August 2026, with encounter number, age, ward, hospital, prior admissions and predicted risk.

#	Element	Problem it detects	Decision supported
1	Status tiles	Which headline measure is off target	Where to look first
2	Hospital D control chart	Sustained rise above baseline	Escalate and investigate

#	Element	Problem it detects	Decision supported
3	Follow-up lines with target	Follow-up collapse in one hospital	Resource follow-up team
4	Bars with and without follow-up	Size of the benefit of follow-up	Make the case for investment
5	Priority call list	Highest-risk discharges to contact	Daily call workload; clinician decides

6.3 One-Page Briefing in Words

Zone	Content
Headline	Network readmissions are steady at 14.5%, but Hospital D has risen from 15.0% to 18.0% as follow-up fell from 56% to 30%.
Evidence	Hospital D control chart; two bars comparing readmission with and without follow-up
Decision requested	Approve staffing for the Hospital D follow-up team within 3 months
Data note	Source, definitions, refresh date, small-number suppression, contact

6.4 Build Plan: Tools and a Model Sketch

The tool choice from the visualization plan is applied: PostgreSQL stores and models the data, Python prepares it and produces statistical charts, and Power BI delivers the dashboard. A simple star schema (fact table of discharges; date, hospital and ward dimensions) lets every measure be defined once. Illustrative measures:

```
Readmission Rate = DIVIDE( SUM( Fact_Discharge[Readmit30] ), COUNTROWS( Fact_Discharge ) )
Follow-up Rate   = DIVIDE( SUM( Fact_Discharge[Followup7] ), COUNTROWS( Fact_Discharge ) )
Baseline Rate    = CALCULATE( [Readmission Rate], Dim_Date[MonthIndex] <= 11, ALL( Dim_Date
) )
Upper Limit      = VAR p = [Baseline Rate]  VAR n = COUNTROWS( Fact_Discharge )
                   RETURN p + 3 * SQRT( DIVIDE( p * ( 1 - p ), n ) )
```

6.5 Accessibility and Plain Language

- Colour-blind-safe palette with shapes and words as well as colour (● on target, ▲ watch, ■ action).
- Alt text and a “Figure in words” description for every chart, so that the content survives without the picture.
- Headline titles state the finding; acronyms are defined; natural frequencies (“20 in 100”) replace odds ratios for general audiences.
- Text at least 12 pt, logical reading order, keyboard-operable filters, data tables available for each chart.

7. Ethics, Privacy and Governance Across the Lifecycle

Stage	Ethical or governance check	Applied in the case
Data strategy	Lawful basis, consent or waiver, ethics approval, licence terms	Simulated data; real use would need approval and a data-use agreement (DPDP Act 2023, ICMR guidance, HIPAA or GDPR as applicable)
Data quality	Minimisation, de-identification, representativeness review	Only fields needed; missing-data patterns by group reviewed

Stage	Ethical or governance check	Applied in the case
Modelling	No unjustified sensitive predictors; leakage control; fairness criterion chosen openly	Rural and income excluded as predictors but used in the audit
Visualization	Small-number suppression, honest axes, accessible design	Counts under 11 suppressed; bars start at zero
Reporting	State limitations and uncertainty; avoid causal overclaims	Section 11; association wording
Deployment and monitoring	Human oversight, accountability, drift and equity review, route to contest decisions	Clinician decides; monthly monitoring; named owner

8. Integration Analysis: Where Strategies Interlink

8.1 Dependencies Between Components

From → To	What passes between them	Example from the case
Data strategy → Quality	Documentation and field definitions define what "correct" means	The follow-up field definition (completed within 7 days) fixed how it could be used
Quality → Analysis	Cleaning decisions change results	Duplicates removed before counting; missing income kept as a group so rural patients were not dropped
Quality → Modelling	Timing and leakage decisions	Follow-up recorded after discharge was kept out of the model
Analysis → Visualization	Uncertainty and structure of results decide the chart	Control chart chosen because hospital months are small and noisy
Visualization → Reporting	Visuals make findings quotable	The two-bar chart supports the "twice as high" message
Reporting → Strategy	Decisions create new data needs	Recommendation to capture out-of-network readmissions
Ethics → all	Constraints on data, variables, display and use	Sensitive attributes used for audit, not prediction

8.2 What Would Have Gone Wrong If Components Were Done in Isolation

Isolated work	Likely failure	Integrated safeguard
Analysis on uncleaned data	153 duplicate encounters would have inflated counts and distorted rates	Quality gate before analysis
Network-level reporting only	Hospital D problem unseen because the network average was flat	Hospital-level visuals chosen at design stage
Modelling without a communication plan	A model with AUROC 0.64 might be read as a decision-maker	Plain-language description, role as aid, calibration shown
Class re-weighting without calibration check	Average predicted risk 47.8% vs 15.0% observed	Calibration measured and re-weighting rejected

Isolated work	Likely failure	Integrated safeguard
Dashboard built before defining decisions	Attractive but unused visuals	Each visual tied to a named decision
Visuals without text descriptions	Content lost for screen-reader users and text-only reviewers	A written description for every figure

8.3 How Integrated Effort Improves Outcomes

- Faster, better-targeted action: combining quality checks, hospital-level analysis and an accessible control chart led directly to one actionable priority (Hospital D follow-up), worth an estimated 37 to 65 fewer readmissions a year.
- Fewer false alarms and missed signals: statistical rules (control limits, run rules) agreed in advance prevent over-reaction to single months.
- Trust: transparency from data source to recommendation lets a sceptical manager retrace each step.
- Equity: linking data completeness, fairness audit and outreach plans means gaps are found and acted on, not only reported.
- Reuse: because the pipeline, definitions and visuals are documented, a new dataset can follow the same route with little rework.

9. Unified Recommendations

#	Recommendation	Evidence	Owner	Timing	Measure of success
O 1	Restore 7-day follow-up at Hospital D (staffing, pre-booked appointments, reminders)	Figures 3, 4	Hospital D nursing director	0–3 months	Follow-up at least 55%; readmission at or below 15%
O 2	Make medication reconciliation standard work, starting at Hospital D	Table 2	Pharmacy lead	0–6 months	At least 85% in every hospital
O 3	Add discharge checklists for weekend discharges and stays over 7 days; evaluate before scaling	Table 2	Ward managers	3–9 months	Lower readmission in those groups
A 1	Report readmissions by hospital and ward with control charts and run rules	Figure 3	Quality and analytics lead	0–2 months	Signals reviewed within one month
A 2	Pilot the risk model in silent mode, then use it to prioritise calls for the top 20%	Section 5.4	Clinical informatics	3–9 months	Contacts reach at least 30% of readmissions
E1	Close the rural and low-income follow-up gap; recalibrate for under-predicted groups	Figure 6	Community health lead	3–12 months	Rural–urban follow-up gap halved
D 1	Apply data-quality gates and a source log; reduce missing income, comorbidity and medication fields	Section 4	Data governance lead	0–6 months	Missing data under 3%
C 1	Deliver the accessible dashboard and one-page briefing with text descriptions	Section 6	Analytics lead	3–6 months	Monthly use by target users; accessibility checklist passed
G 1	Obtain ethics approval; review drift and fairness monthly; keep human decision-making	Section 7	Governance committee	0–12 months	Reviews held; incidents logged

Priorities are ordered by expected impact and urgency: O1 and A1 first, because they address the largest identified problem and make future problems visible; D1 and G1 run alongside as enabling work.

10. Best Practices, Pitfalls and Lessons Learned

10.1 Best-Practice Checklist by Phase

Phase	Best practice	Why it matters
Framing	Write the decision, the people who will act, and the success measure before touching data	Prevents analysis that answers no one's question
Data	Score sources against criteria; log source, version and licence; get approvals first	Traceability and legal safety
Quality	Check five dimensions; record every exclusion and its count; treat missingness as a finding	Trustworthy denominators and fair subgroup results
Analysis	Split by patient and date; fix interpretation rules in advance; report uncertainty	Honest, reproducible results
Modelling	Start with a simple baseline; check calibration, not just ranking; audit subgroups; test pitfalls rather than only naming them	Safe, usable models
Visualization	Title charts with findings; design for accessibility; describe every visual in text	Understanding by all audiences
Reporting	Give worked examples; trace each recommendation to evidence; state limitations	Credibility and actionability
Ethics	Run ethics checks at every stage; keep humans in charge	Protects patients and the organisation
Operations	Monitor drift, quality and equity; assign owners and review dates	Keeps results valid over time

10.2 Pitfalls Encountered and Lessons Learned

Pitfall	What happened	How it was handled	Lesson for next time
Averages hide local problems	The network rate was flat while one hospital worsened	Added hospital-level control charts	Decide the level of analysis from where action can be taken
Class imbalance	Re-weighting looked reasonable but inflated predicted risk to about three times the observed rate	Tested it; kept the plain model with a capacity-based threshold	Always compare calibration before and after any rebalancing
Leakage risk	Follow-up completion is strongly linked to the outcome but happens after discharge	Excluded from the model	Time-stamp every predictor
Weak first simulation	The first simulated dataset produced signals too weak to interpret clearly	Adjusted the design so the mechanisms were detectable, and documented this	State openly that simulated effects were designed; do not present them as discoveries about the real world

Pitfall	What happened	How it was handled	Lesson for next time
Visuals not read as evidence	Reviewers gave no credit for mock-ups shown only as images	Added written descriptions to every figure and mock-up	Assume a reader may see only the text
Abstract methods	Earlier plans named techniques without examples	Added worked examples with calculations	Show one concrete example for every method
Overclaiming causation	Strong follow-up association could be read as proof	Association wording; half-effect scenario	Separate association, prediction and causal claims
No real dataset or hands-on tool	Early tasks had no data and uncertainty about tools	Used a simulated cohort and a staged tool plan	Practise on a public dataset early to build real-data skills

10.3 Recommended Workflow for Similar Projects

1. Write a one-page brief: the decision, the audience, the outcome definition and the success measure.
2. Select and log data sources against the criteria; secure approvals.
3. Run the five-dimension quality checks; produce the cohort flow and risk register.
4. Explore by group and time before modelling; fix interpretation rules in advance.
5. Build a baseline model and one flexible model; evaluate on a locked test period; test class imbalance and leakage explicitly.
6. Audit subgroups and decide on a fairness criterion with stakeholders.
7. Design visuals from decisions; add text descriptions and accessibility checks; test with users.
8. Write the report with worked examples and an evidence-to-recommendation matrix.
9. Pilot with oversight; monitor drift, quality and equity; revise.

A suggested time budget for a project of this size is roughly 20% framing and data, 25% quality and cleaning, 25% analysis and modelling, 15% visualization, and 15% reporting and review. The share spent on quality and cleaning is deliberately high, because every later stage depends on it.

11. Limitations of the Integrated Project

- Simulated data: relationships were designed by the author, and the simulation was adjusted to make mechanisms detectable. The results demonstrate method, not real-world findings.
- Association, not proof: the follow-up effect and the scenario estimates rely on observational patterns and on assumptions about causality.
- Modest model: only routine variables were used; discrimination is modest and the model was not externally validated.
- Single network and outcome definition: readmissions outside the network are not observed.
- Visual and dashboard mock-ups were designed but not built and user-tested in Power BI.
- Small subgroups: fairness findings are signals, with overlapping intervals and no formal multiple-comparison correction.
- Source scoring is judgement-based and the time estimates are planning figures, not measured effort.

12. Conclusion

This project shows that effective healthcare data analysis is a connected process. Choosing and documenting data, checking its quality, analysing it with methods suited to the question, presenting it so that each audience can act, and reporting with transparent evidence each rely on the stage before. In the case, a stable network average hid a rise at Hospital D that was linked to missed follow-up; a modest but well-calibrated model could help prioritise scarce follow-up capacity; and fairness and drift checks showed where the model needs care. The unified recommendations, with owners, timelines and measures, convert these findings into a practical plan. The reflective self-assessment in Appendix A records what was learned, what was weak, and how the next stage of learning will build on it.

Appendix A. Reflective Self-Assessment

A.1 Learning Journey by Stage

Stage	What I produced	What I learned	Challenge	How I responded
1. Data strategy and quality	Plan for public sources, selection criteria, quality framework and risk register	Data must be chosen and checked against explicit criteria; quality has named dimensions	Knowing which datasets exist and under what access conditions	Compared sources by access, documentation and licence; recorded them in a table
2. Visualization	Plan with metrics, tool comparison, chart guide and three dashboard mock-ups	Visuals should start from decisions and users; accessibility is a design requirement	No dataset or familiarity with tools at the start; unsure whether a plan or a built dashboard was needed	Produced a plan with illustrative mock-ups and a staged route to a real prototype
3. Statistics and modelling	Framework covering techniques, pre-processing, model choice, validation and ethics	Leakage, calibration and fairness decide whether a model is usable	Describing methods without numbers made them abstract	Later added worked examples and tested class imbalance
4. Analytical report	Hypothetical case report with simulated data, results and recommendations	A result becomes useful only when linked to a decision and an audience	No real data, so I had to create it carefully and stay honest about it	Built a documented simulation and a fully reproducible script
5. Integration (this report)	Unified report, framework, integration analysis and best practices	The joins between stages are where projects succeed or fail	Keeping five components consistent with one set of numbers	Used one case and one script so all figures agree

A.2 Feedback Received and Improvements Made

Feedback theme	What I changed	Where to see it
Mock-ups of dashboards and reports not recognised as present	Described each mock-up zone by zone in words, with example wording	Sections 6.2, 6.3; “Figure in words” notes
Need concrete examples and specifics	Added worked examples with calculations and real case numbers	Sections 4.2, 5, 6.4; companion analytical report Section 3.9

Feedback theme	What I changed	Where to see it
Practical solutions for class imbalance	Compared remedies and tested class re-weighting on the data	Section 5.4; Figure 5; Section 10.2
A case study illustrating the whole process	Ran one case from data strategy to recommendations	Sections 4 to 9
Accessibility explanations needed more detail	Listed specific rules, a palette, and alt-text practice	Section 6.5

A.3 Critical Assessment of My Work

Strengths

- Consistent structure: each stage links to a decision, and every recommendation is traceable to evidence.
- Reproducibility: the simulation, cleaning, analysis and figures run from one script with fixed seeds.
- Honesty about limits: the simulated nature of the data, the association-versus-cause distinction and the modest model are stated openly.
- Attention to ethics and accessibility at every stage rather than at the end.

Weaknesses

- No real, messy dataset has yet been analysed, so my data-cleaning judgement is untested on real-world problems.
- The dashboard exists as a design and a description, not a built and user-tested Power BI report.
- The model is simple; I have limited experience tuning and comparing more advanced models or using causal methods.
- The early plans were too abstract; I improved them only after receiving feedback, and should have added examples from the start.

What I would do differently

- Begin with a small public dataset, such as the UCI diabetes readmission data, and build a real dashboard before writing the plan.
- Test every named pitfall on data as soon as it is mentioned.
- Seek feedback from a user or mentor after each draft, not only at the end.

A.4 Skills Inventory with Evidence

Skill	Evidence in this project	Current level	Next step
Data sourcing and quality assessment	Source scoring, five-dimension checks, risk register	Working	Apply to a real public dataset with genuine errors
Python (pandas, scikit-learn, statsmodels)	Cleaning, regression, modelling and figure scripts	Working	Write more code independently and review it with others
Statistics and modelling	Control charts, logistic regression, calibration, fairness audit	Working	Strengthen theory; learn causal inference basics
SQL and databases	Designed a star schema; not yet hands-on in this project	Developing	Practise joins and window functions; load the cohort to PostgreSQL

Skill	Evidence in this project	Current level	Next step
Power BI and DAX	Mock-ups and measure sketches	Developing	Build and publish the dashboard
Data visualization and accessibility	Chart guide, palette, alt text, text descriptions	Working	User-test with real stakeholders
Report writing and communication	Executive summaries, plain-language messages, matrices	Working	Present findings orally to a non-technical audience
Ethics and governance	Lifecycle ethics table, fairness choices	Working	Study applicable laws and ethics review processes in more depth

Levels are a personal judgement: Developing = understand the idea, little hands-on use; Working = can apply with guidance and checking.

A.5 Challenges Faced and How They Were Overcome

- Starting without data or tool experience: solved by planning first, then simulating data and scripting the work.
- Making results believable without real data: solved by a transparent simulation, fixed seeds and a statement of what was designed.
- Turning technical output into messages for managers: solved by headline titles, plain-language statements and “out of 100” wording.
- Keeping five documents consistent: solved by a single case and one script that produces every number.

A.6 Career Development Pathway

The goal is to become a healthcare data analyst who can take a question from raw data to an accessible, ethical recommendation. My background in sociology is an asset for health equity, social determinants and survey-based evidence, and the pathway builds technical depth on top of it. The targets below are proposed and will be reviewed every three months.

Horizon	Goals	Evidence of progress
Next 6 months	Publish four portfolio repositories with clear READMEs; build the Hospital D dashboard in Power BI on a real public dataset; complete SQL practice (joins, window functions); prepare for a recognised analytics credential such as the Microsoft Power BI Data Analyst certification; apply for junior data analyst or healthcare analytics roles each month	Repositories online; .pbix file and screenshots; completed SQL exercises; certification date; application log
6–18 months	Complete a real-data health analytics project with documented ethics considerations; learn causal inference and model-monitoring basics; learn health data standards (for example ICD-10 coding and FHIR); give a presentation of findings to a non-technical audience	Project report; short course certificates; presentation recording or slides
2–3 years	Move to a healthcare or public-health analyst role, then toward senior analyst or health data scientist responsibilities; mentor newcomers; contribute to open health-data projects	Role progression; reviewed contributions; feedback from managers

Specific goal	Measure	Target date
Four portfolio repositories published	Four public repos with README, data dictionary and licence	January 2027
Real-data Power BI dashboard	Published .pbix and screenshots in a repository	December 2026

Specific goal	Measure	Target date
SQL competence	30 practice queries completed and the cohort loaded to PostgreSQL	January 2027
Analytics credential	Exam passed	March 2027
Job search	At least 10 relevant applications per month	From December 2026

Dates and credentials are proposals to be adjusted to circumstances and job-market feedback.

A.7 Commitments for Improvement

1. Add a worked example and a text description to every future method or visual from the first draft.
2. Practise on real, messy public data at least once a month and log the cleaning decisions.
3. Ask a mentor or peer to review each major deliverable before submission and record the feedback and my response.

Appendix B. Project Work Breakdown

Task	Hours
Reviewing earlier outputs and extracting key insights (K1 to K8)	4.0
Designing the unified structure and the framework diagram	3.0
Re-running the case, quality checks and building integrated figures and dashboard	5.0
Writing the integrated sections (Sections 4 to 7)	9.0
Integration analysis: interlinks, isolated-failure analysis, unified recommendations	3.0
Best practices, pitfalls and lessons learned	3.0
Self-assessment and career pathway	4.0
Formatting, accessibility text, referencing and quality review	2.0

Total estimated effort: approximately 33 hours.

Appendix C. Component Cross-Reference and Repository Index

Component	Document	Where to find it
Data collection and quality strategy	Healthcare Data Collection and Quality Strategy (Word)	Capstone repository, docs folder
Data visualization plan	Healthcare Data Visualization Plan (Word)	Visualization repository (link below)
Statistical analysis and predictive modeling framework	Healthcare Predictive Modeling Framework (Word)	Capstone repository, docs folder
Analytical report with simulated case	Healthcare Analytical Report (Word), code, data	Analytics repository (link below)
Integrated report (this document)	Integrated Project Report (Word)	Capstone repository (link below)

Capstone: https://github.com/ipshitav71-wq/healthcare-data-analysis-capstone
Visualization plan: https://github.com/ipshitav71-wq/healthcare-data-visualization-plan
Analytical report and code: https://github.com/ipshitav71-wq/healthcare-analytics-report-simulated

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